PABLO VILLANUEVA DOMINGO

PHD IN PHYSICS & DEEP LEARNING SCIENTIST

AI scientist specializing in **deep learning** and **multi-agent systems**, with a **PhD in theoretical physics** (*cum laude*, University of València). Currently developing RL pipelines and autonomous agents for the CARLA simulator (UAB/CVC) and scientific discovery platforms. Background in applying deep learning to cosmology and astrophysics; 23 publications, h-index 17, 1300+ citations.

CONTACT

- Pablo.Villanueva.Domingo@gmail.com
- pablovd.github.io
- @PabloVD
- pablo-villanueva-domingo-76b6491b2
- 0000-0002-0936-4279
- Pablo Villanueva Domingo
- @CosmoPabloVD

SKILLS

Computation

Programming languages
Python, Typescript, C++, C, C#, Fortran, SQL
Web/App development
React, React Native, Tailwind, Jekyll, FastAPI
General software
Git, Docker, LaTeX, MATLAB, Mathematica
Data analysis
Numpy, SciPy, Pandas, OpenCV, Networkx
Visualization
Matplotlib, Seaborn, Plotly, Folium
Simulation & game engines
Unreal, Godot, Chrono, Flex, Unity
Autonomous driving & robotics software
CARLA, IsaacLab, ROS, Scenic, Xviz

Machine learning

ML libraries
PyTorch, TensorFlow, PyTorch Lightning, PyTorch Geometric, Scikit-learn
Neural Nets Architectures
Graph (GNNs), Convolutional (CNNs), U-Nets, Diffusion models, GANs, LSTMs
Agents & LLMs
Langchain/Langgraph, AG2, Denario, RAGs
Fields
Computer vision, Natural Language Processing, Reinforcement Learning
See my work in ML and programming at
https://pablovd.github.io/codes

Soft skills

Communication
Public speaking, writing skills
Project management
Collaboration, teamwork, initiative, organization
Problem solving
Logical reasoning, lateral thinking, creativity, data modeling

Languages

Spanish	Mother tongue
Catalan	Mother tongue
English	Fluent
Portuguese	Basics

WORK HISTORY

- Deep Learning Scientist

Jan. 2022- Now | Computer Vision Center - Universitat Autònoma de Barcelona

Reinforcement learning training, Data-driven traffic models, neural terramechanics and computer vision tasks at the autonomous driving simulator CARLA

- Research assistant

Jun. 2021- Dec. 2021 | Instituto de Física Corpuscular - Universitat de València

Técnico superior de apoyo a la investigación, CIDEGENT/2018/019, CPI-21-108

- PhD fellowship

Oct 2016 - Mar. 2021 | Instituto de Física Corpuscular - Universitat de València

FPI Severo Ochoa, Ref. SEV-2014-0398-16-3

- Research introduction fellowship

May-Oct. 2016 | Instituto de Física Corpuscular

Iniciación a la investigación Severo Ochoa

MAJOR PROJECTS

- Denario – Multi-agent AI Scientist Platform

2024 – Now | astropilot-ai.github.io/DenarioPaperPage

Co-creator of a multi-agent system for AI assistance of research and end-to-end scientific discovery, spanning 12 disciplines. 36 co-authors and 500+ GitHub stars.

- CARLA – Autonomous Driving Simulator

Jan. 2022 – Now | carla.org

Development of machine learning pipelines, data-driven traffic models, and computer vision modules for the open-source CARLA simulator.

- AeroSim – Aviation Simulation Platform

2024 – Now | github.com/aerosim-open/aerosim

Development of the open-source AeroSim simulation platform and its visual user interface for aerospace applications.

EDUCATION

- PhD in Physics, *cum laude*

2016-2021 | Instituto de Física Corpuscular - Universitat de València

- Master in Advanced Physics

2015-2016 | Universitat de València

- Bachelor of Physics

2011-2015 | Universitat de València

As well as multiple PhD schools and courses which can be found here.

AWARDS

Feb. 2023 | CSIC 2021 relevant PhD Thesis Award, by Consejo Superior de Investigaciones Científicas (CSIC).

Dec. 2016 | 1st prize in the *XXVII edición del Premio Rotary al Fomento del Trabajo Experimental en Física*.

RESEARCH STAYS

I have participated in several international research collaborations, visiting universities from different countries:

Nov.- Dec. 2019 | 3 weeks at Service de Physique Théorique, Université Libre de Bruxelles, Brussels, Belgium.

Sep.- Oct. 2019 | 1 month at Department of Astrophysical Sciences, Princeton University, New Jersey, USA.

Sep.- Nov. 2018 | 2 months at Kavli IPMU, University of Tokyo, Japan.

Jun.- Aug. 2017 | 2 months at Fermi National Accelerator Laboratory (Fermilab), Illinois, USA.

TALKS

I have delivered **9 invited seminars** at the universities of Princeton (USA), Tokyo, Nagoya (Japan), Brussels and València, and presented **8 talks** at international conferences, meetings and schools.
A complete list can be found at https://pablovd.github.io/talks.pdf. These are some selected talks:

• *Neural Terramechanics and the RACER-SIM project*
Jul. 28 2022 | EAI Tech Forum, Intel Labs, online

• *Weighing the Milky Way with AI*
Jan. 17 2022 | *Cosmology Talks*, online (Youtube channel) | Video

• *Machine Learning at galactic and cosmological scales*
Nov. 17 2021 | Instituto de Física Corpuscular | Video and slides

SELECTED PUBLICATIONS

I have published **23 scientific articles** in high impact journals based on my research on cosmology, deep learning and multi agent systems, with over **1300 citations** and an **h-index of 17** (Google Scholar). The full list of publications can be found in my INSPIRE profile P.Villanueva.Domingo.1. These are some selected publications:

• *The Denario project: Deep knowledge AI agents for scientific discovery*
Francisco Villaescusa-Navarro, Boris Bolliet, **Pablo Villanueva-Domingo**, Adrian E. Bayer, et al.
Oct. 2025 | arXiv:2510.26887
A multi-agent AI system for assistance in scientific research, spanning multiple fields from physics to medicine.

• *Weighing the Milky Way and Andromeda with Artificial Intelligence*
Pablo Villanueva-Domingo, Francisco Villaescusa-Navarro, Shy Genel, Daniel Anglés-Alcázar, Lars Hernquist, Federico Marinacci, David N. Spergel, Mark Vogelsberger and Desika Narayanan
Nov. 2021 | Physical Review D 107, 103003, 2023, arXiv:2111.14874
The total masses of the Milky Way and Andromeda galaxies are predicted using AI for the first time, via Graph Neural Networks.

• *Removing Astrophysics in 21 cm maps with Neural Networks*
Pablo Villanueva-Domingo and Francisco Villaescusa-Navarro
Jan. 2021 | The Astrophysical Journal, 907(1):44, 2021; arXiv:2006.14305
The cosmic density field is predicted from maps of distribution of hydrogen training a U-Net in PyTorch.

OUTREACH & ADDITIONAL WORK EXPERIENCE

Feb. 2021 | Outreach video about the astronomer Sandra M. Faber within the project *Pioneras - Recordando a Lise Meitner*.

2020 - Now | Journal referee for journals such as MNRAS and ApJ. See reviews in my Publons profile

Jun. 2019 | Member of the local organizing committee of the Invisibles19 Workshop at València and Invisibles19 School at Laboratorio subterráneo de Canfranc (LSC)

2016-2017 | Collaboration in the organization of the outreach event *Feria-Concurso Experimenta*, València.